# 9 - The Periodic Table

19/12/24

## Periodicity of physical properties of the elements in Period 3

⇒ atomic radius

- decreases gradually, because nuclear charge increases, thus outer electrons are more attracted towards the nucleus, making the atom smaller

⇒ ionic radius

- cations (Na, Mg, Al, Si) are smaller than their respective atoms because by giving electrons, they lose a shell. The remaining electrons are more strongly attracted to the same nuclear charge
- anions (P, S, Cl) are bigger than their respective atoms because they have more electrons than protons so less attraction to the nucleus. The added electrons also create a repulsion which expands the ion

[anions are bigger than cations because they one more shell]

⇒ melting point

- increases from Na → Al because the strength of the metallic bond increases
- Si has the highest MP because it has a giant covalent structure, requiring high energy to overcome
- P, S, Cl, Ar have a lower MP because their simple molecular structures have weak van der Waal's forces

S < P < Cl < Ar because these elements exist as $S_8$, $P_4$, $Cl_2$, Ar. $S_8$ has the most electrons thus stronger van der Waal's forces

⇒ electrical conductivity

- Na → Al are metals, Si is a semi-metal & P → Ar are non-metals
- elec. conductivity is highest in metals, lower in semi-metals & zero in non-metals (most are insulators)
- depends on number of electrons contributed by per atom to the sea of delocalised electrons. Increases from Na → Al, Si is a semi-conductor, P → Ar do not have free mobile electrons to carry the electricity

## Reactions with oxygen

$4Na(s) + O_2(g) \rightarrow \underbrace{2Na_2O(s)}_{\text{white sodium oxide}}$

→ burns with an orange-yellow flame

$2Mg(s) + O_2(g) \rightarrow \underbrace{2MgO(s)}_{\text{white magnesium oxide}}$

→ burns with a white flame

$S(s) + O_2(g) \rightarrow \overbrace{SO_2(g)}^{\text{toxic sulfur dioxide}}$

→ burns with a blue flame

→ further oxidation would give sulfur trioxide

$\underbrace{4Al(s)}_{\text{powdered aluminium}} + 3O_2(g) \rightarrow \overbrace{2Al_2O_3(s)}^{\text{white aluminium oxide}}$

→ burns with a white flame

→ when aluminium is exposed to air, it forms an oxide layer to prevent the aluminium from reacting

$P_4(s) + 5O_2(g) \rightarrow \overbrace{P_4O_{10}}^{\text{white phosphorus oxide}}$ (oxygen in excess)

→ burns with a white/yellow flame

→ clouds of white phosphorus oxide form

* silicon has a slow reaction & Chlorine and Argon don't react with oxygen

## Reactions with chlorine

$2Na(s) + Cl_2(g) \rightarrow \underbrace{2NaCl(s)}_{\text{white sodium chloride}}$

→ $Cl_2$ is a green-yellow gas
→ burns with an orange-yellow flame

$Mg(s) + Cl_2(g) \rightarrow \underbrace{MgCl_2(s)}_{\text{white magnesium chloride}}$

→ burns with a white flame

* Sulfur burns in chlorine gas to form disulfur dichloride but we don't need to know that
* Argon does not form a chloride

$P_4(s) + 10Cl_2(g) \rightarrow 4\overbrace{PCl_5}^{\text{yellow phosphorus (v) chloride}}(l)$

(chlorine in excess)

∴ when chlorine is limited, $PCl_3$ forms

$2Al(s) + 3Cl_2(g) \rightarrow \overbrace{2AlCl_3(s)}^{\text{white aluminium chloride}}$

→ burns on heating to form ionic aluminium chloride

$Si(s) + 2Cl_2(g) \rightarrow \underbrace{SiCl_4(l)}_{\text{colorless silicon (IV) chloride}}$

→ burns with a white flame

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| | $Na_2O$ | $MgO$ | $Al_2O_3$ | $P_4O_{10}$ | $SO_2/SO_3$ |
|---|---|---|---|---|---|
| Bonding | ionic | ionic | ionic/covalent (amphoteric) | covalent | covalent |
| Structure | giant ionic | giant ionic | giant ionic | simple molecular | simple molecular |
| ox. number of period 3 element | +1 | +2 | +3 | +5 | +4/+6 |

| | $NaCl$ | $MgCl_2$ | $AlCl_3$ | $SiCl_4$ | $PCl_3/PCl_5$ |
|---|---|---|---|---|---|
| Bonding | ionic | ionic | covalent | covalent | covalent |
| Structure | giant ionic | giant ionic | simple molecular | simple molecular | simple molecular |
| ox. number of period 3 element | +1 | +2 | +3 | +4 | +3/+5 |

## Reactions with water

$2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$

→ sodium catches fire in cold water

→ violent exothermic reaction

$Mg(s) + \overbrace{2H_2O(l)}^{\text{cold water}} \rightarrow Mg(OH)_2(aq) + H_2(g)$

→ reacts slowly, making $H_2$ & a weakly alkaline magnesium hydroxide solution

$Mg(s) + \overbrace{2H_2O(g)}^{\text{steam}} \rightarrow MgO(s) + H_2(g)$

→ reacts rapidly

## Flame colors

| | |
|---|---|
| Na | yellow flame |
| Mg | white flame |
| Al | white flame |
| Si | —— |
| P | yellow-white flame |
| S | blue flame |

⇒ bases are usually metal oxides / hydroxides

| $Na_2O$  $MgO$ | $Al_2O_3$ | $SiO_2$ | $P_4O_{10}$  $SO_2/SO_3$ |
|---|---|---|---|
| ↓ basic oxides (ionic oxides) | ↓ amphoteric oxide | ↓ weakly acidic (inert) | ↓ acidic oxides (covalent oxides) |

| Reaction with water | Acid/Base behaviour |
|---|---|
| $Na_2O_{(s)} + H_2O_{(l)} \rightarrow 2NaOH_{(aq)}$  pH = 13.0<br>→ exothermic reaction producing strongly alkaline solution of NaOH | $\underbrace{Na_2O_{(s)}}_{\text{basic oxide}} + \underbrace{2HCl_{(aq)}}_{\text{acid}} \rightarrow \overbrace{\underbrace{2NaCl_{(aq)}}_{\text{sodium chloride}}}^{\text{salt}} + \overbrace{H_2O_{(l)}}^{\text{water}}$ |
| $MgO_{(s)} + H_2O_{(l)} \rightarrow Mg(OH)_{2\,(aq)}$  pH = 9<br>→ slight reaction (almost insoluble) producing weakly alkaline solution of $Mg(OH)_2$ | $\underbrace{MgO_{(s)}}_{\text{basic oxide}} + \underbrace{2HCl_{(aq)}}_{\text{acid}} \rightarrow \overbrace{\underbrace{MgCl_{2\,(aq)}}_{\text{magnesium chloride}}}^{\text{salt}} + \overbrace{H_2O_{(l)}}^{\text{water}}$ |
| $Al_2O_3$ → insoluble in water<br>✱ can react with acid & base<br>→ doesn't react/dissolve in water due to its high lattice energy | $\underbrace{Al_2O_{3\,(s)}}_{\text{acts as a base}} + \underbrace{6HCl_{(aq)}}_{\text{acid}} \rightarrow \overbrace{\underbrace{2AlCl_{3\,(aq)}}_{\text{aluminium chloride}}}^{\text{salt}} + \overbrace{3H_2O_{(l)}}^{\text{water}}$<br>$\underbrace{Al_2O_{3\,(s)}}_{\text{acts as an acid}} + \underbrace{2NaOH_{(aq)}}_{\text{hot, concentrated alkali}} + 3H_2O_{(l)} \rightarrow \overbrace{\underbrace{2NaAl(OH)_{4\,(aq)}}_{\text{sodium aluminate}}}^{\text{salt}}$ |
| $SiO_2$ → insoluble in water<br>→ doesn't react/dissolve in water due to its strong covalent bonds | $\underbrace{SiO_{2\,(s)}}_{\text{weak acidic oxide}} + \underbrace{2NaOH_{(aq)}}_{\text{alkali}} \rightarrow \overbrace{\underbrace{Na_2SiO_{3\,(aq)}}_{\text{sodium silicate}}}^{\text{salt}} + \overbrace{H_2O_{(l)}}^{\text{water}}$<br>only reacts under extreme conditions (high temp./pressure) |
| $P_4O_{10\,(s)} + 6H_2O_{(l)} \rightarrow 4H_3PO_{4\,(aq)}$  pH = 2<br>→ forms phosphoric (V) acid | $\underbrace{P_4O_{10\,(s)}}_{\text{acidic oxide}} + \underbrace{12NaOH_{(aq)}}_{\text{alkali}} \rightarrow \overbrace{\underbrace{4Na_3PO_{4\,(aq)}}_{\text{sodium phosphate (V)}}}^{\text{salt}} + \overbrace{6H_2O_{(l)}}^{\text{water}}$ |
| $SO_{2\,(g)} + H_2O_{(l)} \rightarrow H_2SO_{3\,(aq)}$  pH = 2<br>→ forms sulfurous acid | $\underbrace{SO_{2\,(g)}}_{\text{acidic oxide}} + \underbrace{2NaOH_{(aq)}}_{\text{alkali}} \rightarrow \overbrace{\underbrace{Na_2SO_{3\,(aq)}}_{\text{sodium (IV) sulfate}}}^{\text{salt}} + \overbrace{H_2O_{(l)}}^{\text{water}}$ |
| $SO_{3\,(g)} + H_2O_{(l)} \rightarrow H_2SO_{4\,(aq)}$  pH = 2<br>→ violent reaction forming sulfuric acid | $\underbrace{SO_{3\,(g)}}_{\text{acidic oxide}} + \underbrace{2NaOH_{(aq)}}_{\text{alkali}} \rightarrow \overbrace{\underbrace{Na_2SO_{4\,(aq)}}_{\text{sodium (VI) sulfate}}}^{\text{salt}} + \overbrace{H_2O_{(l)}}^{\text{water}}$ |

$\boxed{NaCl \quad MgCl_2}$ $\boxed{AlCl_3 \quad SiCl_4 \quad PCl_3/PCl_5}$

↓ ionic chlorides

dissociate in water (no reaction)

↓ covalent chlorides

react with water (hydrolysis reaction)

| Hydrolysis reaction |
|---|
| water molecule breaks down |

$NaCl_{(s)} + aq \longrightarrow Na^+_{(aq)} + Cl^-_{(aq)}$  pH = 7.0

$MgCl_{2(s)} + aq \longrightarrow Mg^{2+}_{(aq)} + 2Cl^-_{(aq)}$  pH = 6.5 − 6.9

observation

: white solid dissolves to form a colorless solution

$Al^{3+}$ $Cl^-$  $H^+$ $OH^-$

$AlCl_{3\,(s)} + 3H_2O_{(l)} \longrightarrow \underbrace{Al(OH)_{3\,(s)}}_{\text{white ppt}} + \underbrace{3HCl_{(g)}}_{\text{white fumes}}$  pH = 3.0 acidic because of HCl

or $Al_2O_3$

$SiCl_{4\,(l)} + 4H_2O_{(l)} \longrightarrow SiO_{2\,(s)} + 4HCl_{(g)}$  pH = 2.0 acidic because of HCl

$PCl_{5\,(s)} + 4H_2O_{(l)} \longrightarrow H_3PO_{4\,(aq)} + 5HCl_{(g)}$  pH = 2.0 acidic because of HCl

observation

: chlorides react with water, giving of white fumes of HCl

**Q] Element X forms a chloride, which reacts with water to form a solution of pH 1. It forms an oxide which has the melting point of 1610°C. The oxide doesn't dissolve in or react with aqueous sodium hydroxide. Deduce the position of element X in The Periodic Table.**

→ its chloride reacts with water to form acidic solution = not group 1/2, has to be group 3-7

→ high M.P of oxide = giant structure & not from period 2 (e.g $CO_2$ is a gas)

→ doesn't dissolve in NaOH = not $Al_2O_3$ ∵ giant covalent structure

Element X is in group 3-7 and period 3 or lower

**Q] Selenium is in group 16 & period 4 of the Periodic Table. Predict some physical and chemical properties of Se.**

→ group 16 = non-metal with a simple molecular structure

Physical ⇒ simple molecular structure : low melting point, doesn't conduct electricity & is insoluble in water

Chemical → reacts with chlorine to form $SeCl_4$, reacts with oxygen to form $SeO_2$

deduced by comparison to sulfur, an element in the same group

*elements in the same group have similar properties